Double-Blind, Placebo-Controlled Pilot Randomized Trial of Methylprednisolone Infusion in Pediatric Acute Respiratory Distress Syndrome

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Objective: Low-dose methylprednisolone therapy in adults with early acute respiratory distress syndrome reduces systemic inflammation, duration of mechanical ventilation, and ICU length of stay. We report a pilot randomized trial of glucocorticoid treatment in early pediatric acute respiratory distress syndrome.

Design: Double-blind, placebo-controlled randomized clinical trial.

Setting: Le Bonheur Children's Hospital, Memphis, TN.

Patients: Children (0–18 yr) with acute respiratory distress syndrome undergoing mechanical ventilation.

Interventions: Patients were randomly assigned to steroid or placebo groups within 72 hours of intubation. IV methylprednisolone administered as loading dose (2 mg/kg) and continuous infusions (1 mg/kg/d) on days 1–7 and then tapered over days 8–14. Both groups were ventilated according to the Acute Respiratory Distress Syndrome Network protocol modified for children. Daily surveillance was performed for adverse effects.

Measurements and Main Results: Thirty-five patients were randomized to the steroid ($n = 17$, no death) and placebo groups ($n = 18$, two deaths). No differences occurred in length of mechanical ventilation, ICU stay, hospital stay, or mortality between the two groups. At baseline, higher plateau pressures ($p = 0.006$) and lower Pediatric Logistic Organ Dysfunction scores ($p = 0.04$) occurred in the steroid group; other characteristics were similar. Despite higher plateau pressures on days 1 ($p = 0.006$) and 2 ($p = 0.025$) due to poorer lung compliance in the steroid group, they had lower $Paco_2$ values on days 2 ($p = 0.009$) and 3 ($p = 0.014$), higher pH values on day 2 ($p = 0.018$), and higher Pao_2/Fio_2 ratios on days 8 ($p = 0.047$) and 9 ($p = 0.002$) compared with the placebo group. Fewer patients in the steroid group required treatment for postextubation stridor ($p = 0.04$) or supplemental oxygen at ICU transfer ($p = 0.012$). Steroid therapy was not associated with detectable adverse effects.

Conclusion: This study demonstrates the feasibility of administering low-dose glucocorticoid therapy and measuring clinically relevant outcomes in pediatric acute respiratory distress syndrome. Changes in oxygenation and/or ventilation are consistent with early acute respiratory distress syndrome pathophysiology and results of similar clinical trials in adults. We propose and design a larger randomized trial to define the role of glucocorticoid therapy in pediatric acute respiratory distress syndrome. (*Pediatr Crit Care Med* 2015; 16:e74–e81)

Key Words: acute respiratory distress syndrome; child; inflammation; lung; mechanical ventilation; steroid

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Many critical illnesses lead to acute respiratory distress syndrome (ARDS) in critically ill children and adults. ARDS-attributable mortality in children (18–27%) is lower than that in adults (27–45%) (1), but it remains higher than most other pediatric conditions (2–4). The prevalence of pediatric ARDS ranges from 2.0 to 12.8 per 100,000 children per year (4, 5), although its true prevalence may be higher (6). Population-based studies have estimated 62,000 pediatric and 2.2 million adult ICU days/year in the United States, associated with 7,700 pediatric and 190,600 adult cases/year and 1,800 pediatric and 74,500 adult deaths/year (4, 5, 7). Although use of positive-pressure ventilation and high oxygen concentrations (hyperoxia) may be lifesaving, their detrimental effects on lung tissue are now undisputed. Other than introduction of low tidal volume (TV) ventilation (8), prone positioning (9–11), or restrictive fluid management (12), an increasing awareness of oxygen toxicity (13–15) and variable results for surfactant therapy (16–19) or neuromuscular blockade (20–23), little progress has occurred in improving the clinical outcomes of ARDS (24). These randomized controlled trials (RCTs) have influenced outcomes in pediatric ARDS based on "assumed" best practices but were mostly performed in adults and extrapolated to children. Systematic reviews of nitric oxide therapy showed no change in clinical outcomes and potential toxicity (25, 26). Use of low-dose steroids in early ARDS appears promising in adults (27–29), but there are no published trials of glucocorticoid therapy in pediatric ARDS.

Systemic inflammatory responses (regulated vs dysregulated) established in early ARDS may determine the subsequent course and outcomes of ARDS (30, 31). Adult patients who fail to improve by day 7 usually manifest persistent elevations of inflammatory cytokines and chemokines in the circulating blood and bronchoalveolar lavage (BAL) fluid (32). In these patients, the early use of steroids (within 72 hr of ARDS) down-regulated systemic inflammation and led to earlier resolution of lung inflammation, resulting in briefer ventilation and shorter ICU length of stay (LOS). The data from seven randomized trials ($n = 593$) investigating prolonged steroid therapy in adult patients with ARDS provide evidence of efficacy, with larger benefits observed with earlier initiation of treatment (< 3 d vs 7 d). When steroids were initiated before day 14, hospital mortality is decreased from 52% to 35% ($p = 0.01$) in early ARDS (33–37), from 38% to 25% ($p = 0.22$) in late ARDS (38, 39), and from 48% to 32% (odds ratio, 0.46; 95% CI, 0.27–076; $p = 0.002$) in the combined groups (GU Meduri, personal communication, 2014).

Clinical trials testing glucocorticoid therapy in ARDS have been reported but only included adult patients. Low-dose glucocorticoid therapy for ARDS in children was documented in a few case reports and a case series of six patients (40–42), where it reduced systemic inflammation and durations of mechanical ventilation (DMV) and ICU stay. Secretory phospholipase A_2 activity and tumor necrosis factor-α were increased in the BAL fluid of infants with ARDS, and these mediators were correlated with greater clinical severity, increased oxygen requirements, and higher ventilator settings (43). Thus, inflammatory mediators are likely to play an important role in the pathophysiology of pediatric ARDS.

The widespread use of steroids in pediatric patients with ARDS has been extrapolated from adult data. This is untenable, as the disease patterns, immunological responses, capacity for lung growth/development, and clinical outcomes of pediatric ARDS are different from those of adult patients. Conditions such as sepsis, pneumonia, aspiration, or trauma can trigger ARDS in children, but there is little information about the susceptibility or risk factors for pediatric ARDS resulting from such triggers. For example, angiotensin-converting enzyme polymorphisms modulate the severity of ARDS in adults (44) but did not influence the prevalence or course of ARDS in 216 children (44, 45); diabetes mellitus may be protective against lung injury in adults, but this has not been reported in children (46). Lung tissue is still developing in children, with ongoing alveolarization and vascular maturation (47). The lungs of children and adults have very different pathophysiological responses to infection and injury (48). Children have lower mortality rates following ARDS as compared with adults (1, 4). These examples and other studies suggest that the pathophysiology, susceptibility, and clinical outcomes of ARDS in children are different from adults.

We designed a randomized, placebo-controlled pilot study to investigate the effects of IV methylprednisolone in children with ARDS, to determine the feasibility and clinical utility of this steroid regimen and the feasibility of measuring clinically relevant outcomes in children with ARDS, and to estimate the magnitude of steroid treatment effects for sample size calculations in future trials. We hypothesized that glucocorticoid therapy initiated within 72 hours of ARDS diagnosis may accelerate disease resolution, improve PaO_2/FiO_2 ratios, and decrease the DMV. The DMV was our primary outcome, whereas secondary outcomes were improvement in PaO_2/FiO_2 ratio, $PaCO_2$, and LOS in the PICU and hospital. Decreased DMV is an accepted measure of improved lung function resulting from decreased lung inflammation, even if overall mortality is only minimally altered by an intervention (24). Many children with ARDS undergo prolonged DMV causing a huge financial burden and high morbidity; therefore, we selected this as our primary outcome variable.

MATERIALS AND METHODS

Following institutional review board approval at University of Tennessee Health Sciences Center and Le Bonheur Children's Hospital and parental permission, patients were enrolled from November 2010 to January 2014. Inclusion criteria were diagnosis of acute lung injury/ARDS (based on the American European Consensus Conference consensus criteria) (49) in children aged 1 month to 18 years who were mechanically ventilated for less than 72 hours. Patients were excluded if they were receiving glucocorticoids, were terminally ill or on hospice care, were immunosuppressed, or had extensive burns, adrenal insufficiency, vasculitis, diffuse alveolar hemorrhage, invasive fungal infection, chronic liver disease, or gastrointestinal bleed within the past 1 month, or conditions with estimated 6-month mortality of 50% or higher. From the literature, we could not estimate the magnitude of study drug effects on our primary outcome in ventilated children with ARDS; therefore, a convenience sample of 35 patients was randomly allocated to treatment or control groups (1:1) using variable blocks.

Interventions

Patients with ARDS were enrolled within the first 72 hours of diagnosis, and the study drug was started soon after enrollment.

Patients in the treatment group received a methylprednisolone loading dose of 2 mg/kg infused over 15 minutes, followed by continuous infusions 1 mg/kg/d on days 1–7, 0.5 mg/kg/d on days 8–10, 0.25 mg/kg/d on days 11 and 12, and 0.125 mg/kg/d on days 13 and 14. Patients in the placebo group received equivalent volumes of 0.9% saline. Patients received the full 14-day steroid regimen if they required mechanical ventilation for 7 days or longer. If patients were extubated between days 1 and 7, they were advanced to day 8 of therapy (methylprednisolone, 0.5 mg/kg/d) and then tapered according to protocol. If patients were discharged from the PICU to the floor, they were continued on the IV taper. Study drug was stopped at 14 days or if the patient was being discharged from the hospital. Steroid therapy protocols that had a positive impact on the DMV in adults were used to design our clinical protocol. Although we argue that the disease patterns, immunological responses, capacity for lung growth/development, and clinical outcomes of pediatric ARDS are different from those of adult patients, we have used these adult studies to design the steroid therapy protocol used in this trial.

Ventilator settings were adjusted in accordance with the ARDS Network (ARDSNet) protocol, modified for children. Briefly, patients were placed on the Servo *i* ventilator (Maquet, Rasttatt, Germany) in a patient-regulated volume control mode, with or without synchronized intermittent mandatory ventilation, and TVs were set at 6–8 mL/kg based on ideal body weight in obese children and actual body weight in nonobese children. Blood gases were obtained as clinically indicated, and permissive hypercapnia was tolerated to a maximum P_{CO_2} of 70 torr, while maintaining blood pH 7.25 or higher. F_{IO_2} was increased to maintain oxygen saturations of 92–94%, while adjusting positive end-expiratory pressure (PEEP) and inspiratory:expiratory ratios according to the ARDSNet protocol to limit their exposure to hyperoxia. The clinical team decided when to use high-frequency oscillatory ventilation (HFOV) or extracorporeal membrane oxygenation (ECMO). Study-related procedures and data collection were continued in the patients transitioned to HFOV or ECMO. To monitor complications associated with glucocorticoid therapy, we performed routine infection surveillance, sent tracheal aspirates for gram stain and cultures every 3–5 days or if clinically indicated, checked blood glucose levels routinely, and took preventive measures (hand hygiene, elevated head of bed, and oral care) to minimize the risk of nosocomial infections or hyperglycemia.

Data Collection

We collected data on demographics and clinical diagnoses at study entry; type, mode, and duration of mechanical ventilation; Pediatric Risk of Mortality (PRISM) III, Pediatric Index of Mortality 2, and Pediatric Logistic Organ Dysfunction (PELOD) scores for severity of illness; LOS in the PICU and hospital; peak inspiratory, plateau, end-expiratory, and mean ventilator pressures; TVs; respiratory rates; blood gases; and number of lobes affected on chest radiograph were recorded at baseline, which was on day 1 as soon as patient was enrolled and then daily until study exit. The daily data were collected from the electronic medical record, the 4 AM ventilator settings were recorded, and the highest pH and lowest P_{CO_2} values of the each 24-hour period were recorded. For obese patients, TV was set based on ideal body weight. Derived variables included the Pa_{O_2}/F_{IO_2} ratio, oxygenation index (OI), and ventilator-free days at 28 days. Primary investigators who were blinded to the study groups examined the chest radiographs and reviewed radiology reports to confirm the number of lobes affected.

Statistical Analyses

We used GraphPad Instat (version 3.10; Graph Pad Software, San Diego, CA) for blinded statistical analyses to compare the DMV, Pa_{O_2}/F_{IO_2} ratios, LOS, and other outcomes between patients receiving IV methylprednisolone versus placebo. For data distributed normally or otherwise, the randomized groups were compared in intention-to-treat analyses using Student *t* tests or Mann-Whitney *U* tests, respectively. Repeated measures over time were analyzed using analysis of variance (ANOVA) or Kruskal-Wallis ANOVA as appropriate. *p* Value of less than 0.05 was considered statistically significant for these pilot study data analyses.

RESULTS

Based on the Consolidated Standards of Reporting Trials guidelines (50), we screened 91 patients for this pilot study and 23 did not meet our inclusion criteria. Of the remaining 68 patients, 20 patients (29%) met the exclusion criteria and the parents/guardians of 13 patients (19%) refused permission for study participation (**Fig. 1**). The most common reason for exclusion was that patients were already receiving glucocorticoids at the time of screening. We enrolled 35 patients and all patients had complete follow-up. Two patients died in the placebo group, whereas all survived in the steroid group.

At baseline, the placebo group had significantly lower plateau pressures ($p = 0.006$) and higher PELOD scores ($p = 0.04$) as compared with the steroid group. No other differences occurred between the two groups (**Table 1**). Both groups had similar extent of lung injury (Pa_{O_2}/F_{IO_2} ratios and number of lobes affected on chest radiograph) and were managed using similar ventilator modes, TVs, mean airway pressures, and PEEP.

We found no differences in the DMV, our primary outcome measure (**Fig. 2*A***) ($p = 0.94$). Significant improvements occurred in the steroid group Pa_{O_2}/F_{IO_2} ratios on day 8 ($p = 0.047$) and day 9 ($p = 0.002$) with higher means compared with the placebo group (**Fig. 2*B***). The plateau pressures were higher on day 1 ($p = 0.006$) and day 2 ($p = 0.025$), indicating poorer lung compliance in the steroid group as compared with the placebo group (**Fig. 2*C***). Despite this difference, Pa_{CO_2} values in the steroid group were lower on day 2 ($p = 0.009$) and day 3 ($p = 0.014$) and blood pH was higher on day 2 ($p = 0.018$) compared with the placebo group, but no differences occurred on other days (**Fig. 2**, ***D*** and ***E***).

No differences occurred in the ventilator-free days, DMV, OI, and LOS in the PICU or hospital (**Table 2**). Fewer patients in the steroid group required nebulized racemic epinephrine for postextubation stridor ($p = 0.04$) or supplemental oxygen

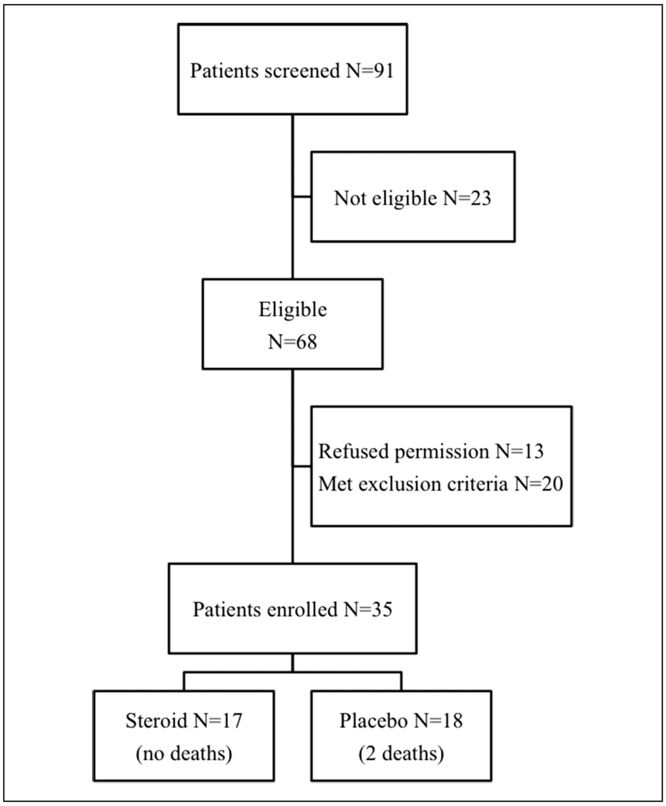

Figure 1. Flow diagram showing the screening, enrollment, and follow-up of patients according to the Consolidated Standards of Reporting Trials guidelines.

at transfer from the PICU ($p = 0.012$). Racemic epinephrine nebulizer was ordered by the PICU attending physicians, based on patient's clinical condition after extubation. Hypertension developed in one patient in the steroid group during the study, and the study drug was discontinued on day 3. The hypertension resolved subsequently, and the patient was discharged home without complications. Tracheal aspirates were checked as regular surveillance cultures every 3–5 days or if clinically indicated. Routine BAL was not performed as surveillance for ventilator-associated pneumonia, as this is not standard of care for pediatric patients currently. Steroid therapy did not increase the risks of hyperglycemia or nosocomial infections between the two groups.

DISCUSSION

We present a pilot randomized trial comparing glucocorticoid versus placebo infusions in early pediatric ARDS. This pilot trial was designed to determine the feasibility and clinical utility of the steroid regimen chosen and the feasibility of measuring clinically relevant outcomes in children with ARDS and to estimate the steroid treatment effects on these clinical outcomes.

At study enrollment, the placebo group had higher PELOD scores than the steroid group, but there were no differences in the risk of mortality predicted by these scores. Two critically ill patients skewed the data for PELOD scores in the placebo group, and these patients died later in their hospital course. These differences were probably due to a small sample size and two outliers in the placebo group who died later. In a larger RCT, stratifying patients based on their PELOD or PRISM III scores may be considered to see if steroid therapy is more (or less) effective in the sicker patients with ARDS.

Consistent with earlier adult studies (27) and reported cases in pediatric patients with ARDS (40–42), we found higher $Pa{O_2}/F{IO_2}$ ratios in the steroid group on days 8 and 9 (after 7 d of steroid infusion) and fewer patients required supplemental oxygen at PICU transfer. We speculate that steroid therapy may have reduced lung inflammation, and we have launched efforts to examine inflammatory markers in these patients. Unlike the adult clinical trials or pediatric cases reported (GU Meduri, personal communication, 2014) (40–42), these differences did not decrease the DMV, ventilator-free days, and PICU or hospital LOS. Higher plateau pressures occurred in the steroid group at enrollment, suggesting poorer compliance than placebo group, but lower $Pa{CO_2}$ values on days 2 and 3 and higher pH on day 2 in the steroid group could indicate a decrease in dead space ventilation (51). Similar to a recent meta-analysis (52), methylprednisolone infusions appeared to be well tolerated by the study subjects, although the safety of this regimen cannot be established from the small number of patients in this pilot study.

A meta-analysis of five randomized trials using glucocorticoids in adult patients with ARDS ($n = 518$) showed improvements in gas exchange, reduction in markers of inflammation, and decreased durations of mechanical ventilation and ICU stay, but a distinct survival benefit only occurred if therapy was initiated in early ARDS (28–30). Our glucocorticoid regimen was designed to replicate the positive clinical trials in adult patients with ARDS. This seemed as good a place to start as any other. We selected patients with early ARDS, used low-dose methylprednisolone, infused it continuously for 7 days, and then tapered it slowly over the next 7 days. Thus, we tried to avoid prolonged ventilation due to muscle weakness, resulting from extended steroid exposure (53), the side effects of high-dose or bolus-dose methylprednisolone (54), and the potential for rebound inflammation by sudden termination of steroid therapy (39).

The strengths of this study include enrollment of a generalizable PICU population, a randomized placebo-controlled study design, use of masked study drug infusions, standardized regimen of low-dose methylprednisolone, a priori definition of primary and secondary outcomes as well as adverse drug effects, complete follow-up of all enrolled subjects, and an intention-to-treat analysis. The results suggest the feasibility delivering our steroid regimen and the feasibility of measuring predefined outcomes in children with ARDS, although no differences occurred in the clinical outcomes or process outcomes between the two randomized groups.

This study has several limitations. Being a pilot study, a sample size was not calculated. Given the absence of any published data from clinical trials in pediatric patients, we had to design an exploratory study to determine the feasibility and to estimate the potential magnitude of effects associated

TABLE 1. Patient Characteristics at Study Entry

Characteristics	Steroid ($n = 17$)	Placebo ($n = 18$)	p
Age (mo)	98.6 ± 82.8	46.4 ± 49.5	0.26
Height (cm)	109.5 ± 44.8	88.5 ± 26.0	0.20
Weight (kg)	33.3 ± 29.8	16.4 ± 9.3	0.21
Male gender (%)	7/17 (41)	11/18 (61)	0.25
African-Americans (%)	11/17 (65)	10/18 (56)	0. 73
Pediatric Risk of Mortality III score	10.8 ± 6.7	12.3 ± 7.7	0.54
Pediatric Index of Mortality-2 score	6.9 ± 8.3	12.6 ± 13.4	0.14
Pediatric Logistic Organ Dysfunction score	7.0 ± 8.5	13.6 ± 9.4	**0.04**
Pao_2/Fio_2 ratio	156.3 ± 57.5	158.1 ± 87.3	0.94
Positive end-expiratory pressure	7.5 ± 2.4	7.9 ± 2.6	0.62
Plateau pressure	33.2 ± 12.6	24.4 ± 7.1	**0.006**
Tidal volume (mL/kg)	6.1 ± 1.6	6.4 ± 2.0	0.62
Mean airway pressure	16.8 ± 6.0	14.4 ± 6.1	0.24
Blood gas pH	7.42 ± 0.09	7.39 ± 0.06	0.22
$Paco_2$	43.8 ± 21.2	40.8 ± 10.9	0.80
No. of lobes affected on chest radiograph	4.0 ± 0.9	3.8 ± 0.9	0.59
Conditions associated with ALI/acute respiratory distress syndrome (%)			0.27
Pneumonia	7/17 (41)	12/18 (67)	
Bronchiolitis	4/17 (24)	3/18 (17)	
Aspiration	5/17 (29)	7/18 (39)	
Trauma	2/17 (12)	0/18 (0)	
Transfusion-related ALI	0/17 (0)	1/18 (6)	
Near drowning	2/17 (12)	2/18 (11)	
Hydrocarbon ingestion	0/17 (0)	3/18 (17)	
Preterm birth	3/17 (18)	2/18 (11)	
Asthma	1/17 (6)	1/18 (6)	

ALI = acute lung injury.

Values presented as mean ± SD. Randomized groups were compared using the two-tailed unpaired t tests for parametric data and the Mann-Whitney U test for nonparametric data. Boldface values are statistically significant.

with steroid therapy. These pilot data will inform the design and power analysis for a definitive study to test the efficacy of steroids in pediatric ARDS. This dose and duration of steroid therapy could be associated with steroid-induced myopathy or increased nosocomial infections, but we did not enroll sufficient numbers to support the safety of this regimen. However, we closely monitored study subjects for the development of these and other complications, but we did not find any differences between the two groups. Due to the small sample size, we also found differences in diagnoses and severity of illness between the two groups. Additional limitations include our failure to strictly control the ventilator management and failure to standardize the ventilator weaning protocol or test for extubation readiness in the enrolled patients. Given the strict masking of study drug infusions, however, we can safely assume that clinical bias for or against steroid therapy could not have affected the clinical decision making around ventilator management, weaning, or extubation.

We speculate that 25–30% reduction in the DMV in the steroid versus placebo group would be both clinically meaningful and achievable based on adult data. Pooled data from pediatric ARDS studies showed an average DMV of 12.5 ± 9.6 days (mean ± SD) (1, 55, 56). Power calculations were based on α-error of p value of less than 0.01, with a threshold of 80% power (1 – β error). Using a two-sided z test and adjusting for one interim analysis after 50% enrollment of patients (using an αt^2 α-spending function), samples sizes for the final intention-to-treat analyses are listed in **Table 3**. To adjust for deaths before extubation and loss to follow-up, the total sample size ($N = N1 + N2$) will be rounded up to 300 subjects. While using explicitly defined protocols for ventilator management and other PICU therapies, we propose stratified randomization by 1) the participating site; 2) the subject's age, categorized as 1–24 months (infant), 2–10 years (preadolescent), and 11–18 years (adolescent); and 3) pulmonary versus extrapulmonary ARDS. Study outcomes will be chosen to collect quantitative, qualitative, and mechanistic data for ARDS in infants and children, examining the roles of genetic, demographic, and clinical risk factors contributing to the pathophysiology of pediatric ARDS.

This pilot study addresses a vital question, namely, is there a role for steroid therapy in pediatric ARDS? This question has been hitherto overlooked, despite its importance for large numbers of mechanically ventilated pediatric patients, as well as the widespread and variable use of steroids in this population. Our study was designed to investigate the effects of IV methylprednisolone in pediatric ARDS, to determine the feasibility and utility of this steroid regimen, to determine the feasibility of measuring

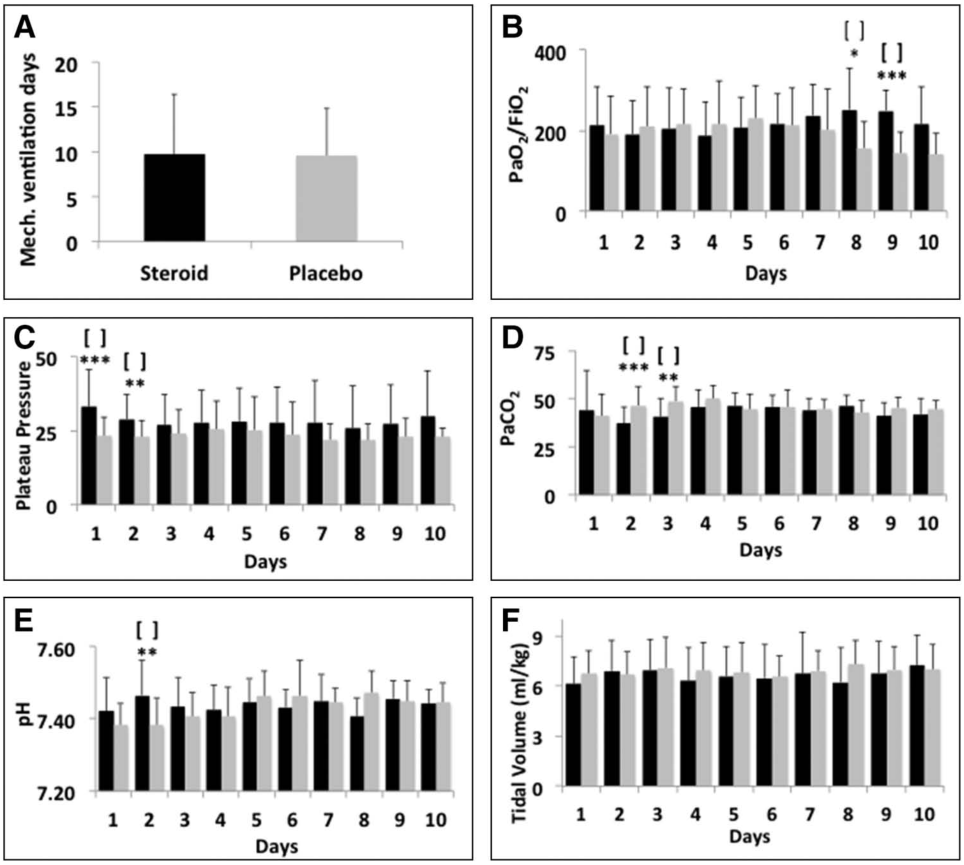

Figure 2. Comparison of steroid group (*black bars*) and placebo group (*gray bars*). **A**, Duration of mechanical ventilation in days. **B**, Pao_2/Fio_2 ratios on days 1–10 ($^*p = 0.047$; $^{***}p = 0.002$). **C**, Plateau pressures days 1–10 ($^{***}p = 0.006$; $^{**}p = 0.025$). **D**, Highest $Paco_2$ on days 1–10 ($^{***}p = 0.009$; $^{**}p = 0.014$). **E**, Lowest pH on days 1–10 ($^{**}p = 0.018$). **F**, Tidal volumes on days 1–10. The numbers of patients assessed on each study days 1–10 for the steroid and placebo groups are listed below.

Study days	1	2	3	4	5	6	7	8	9	10
Steroid group (*n*)	17	16	15	13	12	11	10	10	9	5
Placebo group (*n*)	16	16	16	14	13	10	9	7	6	5

TABLE 2. Clinical Outcomes at Study Exit

Characteristics	Steroid Group (n =17)	Placebo Group (n = 18)	p
Ventilator-free days	17.65 ± 6.74	16.33 ± 7.27	0.58
Duration of mechanical ventilation, d	9.74 ± 6.62	9.59 ± 5.21	0.94
Length of PICU stay, d	13.47 ± 6.63	15.17 ± 8.26	0.51
Length of hospital stay, d	19.35 ± 9.14	26.22 ± 15.68	0.24
Survival to hospital discharge	17 (100%)	16 (88%)	0.15
Racemic epinephrine on extubation	2 (12%)	7 (44%)	**0.04**
Supplemental O_2 at transfer from PICU	13 (76%)	16 (100%)	**0.01**
Hyperglycemia[a]	10 (59%)	8 (50%)	0.41
Nosocomial infection[a]	1 (6%)	5 (31%)	0.09

[a]Hyperglycemia was defined as blood glucose > 150 mg/dL; nosocomial infections were defined based on 2010 Centers for Disease Control and Prevention criteria; tracheal aspirates were sent for culture every 3–5 d as an infection surveillance.

Values are presented as mean ± SD or number of patients (%). Randomized groups were compared using the two-tailed unpaired *t* tests for parametric data, Mann-Whitney *U* tests for nonparametric data, and Fisher exact tests for events. Boldface values are statistically significant.

TABLE 3. Proposed Power Analyses Based on Duration of Mechanical Ventilation as the Primary Outcome

DMV in Placebo Group	DMV in Steroid Group	Percent Reduction in DMV (%)	α-Error	Power (1 – β Error) (%)	No. of Subjects (*N*1)	No. of Subjects (*N*2)
12.5 ± 9.6	10.6 ± 8.5	15	0.01	80.0	547	547
12.5 ± 9.6	9.0 ± 6.9	28	0.01	80.0	137	137
12.5 ± 9.6	8.0 ± 8.5	36	0.01	80.3	98	98

DMV = duration of mechanical ventilation.

All values = mean ± SD. Power calculations were based on an α-error of $p < 0.01$, a threshold of 80% power (1 – β error), and a two-sided z test with adjustment for one interim analysis after 50% enrollment of patients (using an αt^2 α-spending function).

clinically relevant outcomes in pediatric ARDS, and to estimate the magnitude of treatment effects for sample size calculations in future trials (55). Despite the limitations described above, we have accomplished all these objectives. In conclusion, this pilot study demonstrates the feasibility of using steroids and measuring clinically relevant outcomes in children with ARDS. These results support the need for a larger, pivotal RCT, as they are consistent with the pathophysiology of early ARDS in children and the results of similar clinical trials in adult patients with ARDS. A larger trial may also identify subgroups of patients with pediatric ARDS who may benefit from steroid treatment.

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REFERENCES

1. Khemani RG, Smith LS, Zimmerman JJ, et al: Pediatric acute respiratory distress syndrome (ARDS): Definition, incidence and epidemiology. *Pediatr Crit Care Med* 2014; In Press
2. Flori HR, Glidden DV, Rutherford GW, et al: Pediatric acute lung injury: Prospective evaluation of risk factors associated with mortality. *Am J Respir Crit Care Med* 2005; 171:995–1001
3. López-Fernández Y, Azagra AM, de la Oliva P, et al; Pediatric Acute Lung Injury Epidemiology and Natural History (PED-ALIEN) Network: Pediatric acute lung injury epidemiology and natural history study: Incidence and outcome of the acute respiratory distress syndrome in children. *Crit Care Med* 2012; 40:3238–3245
4. Zimmerman JJ, Akhtar SR, Caldwell E, et al: Incidence and outcomes of pediatric acute lung injury. *Pediatrics* 2009; 124:87–95
5. Rubenfeld GD, Caldwell E, Peabody E, et al: Incidence and outcomes of acute lung injury. *N Engl J Med* 2005; 353:1685–1693
6. Cely CM, Rojas JT, Maldonado DA, et al: Use of intensive care, mechanical ventilation, both, or neither by patients with acute lung injury. *Crit Care Med* 2010; 38:1126–1134
7. Stapleton RD, Wang BM, Hudson LD, et al: Causes and timing of death in patients with ARDS. *Chest* 2005; 128:525–532
8. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. The Acute Respiratory Distress Syndrome Network. *N Engl J Med* 2000; 342:1301–1308
9. Relvas MS, Silver PC, Sagy M: Prone positioning of pediatric patients with ARDS results in improvement in oxygenation if maintained > 12 h daily. *Chest* 2003; 124:269–274
10. Curley MA, Hibberd PL, Fineman LD, et al: Effect of prone positioning on clinical outcomes in children with acute lung injury: A randomized controlled trial. *JAMA* 2005; 294:229–237
11. Kopterides P, Siempos II, Armaganidis A: Prone positioning in hypoxemic respiratory failure: Meta-analysis of randomized controlled trials. *J Crit Care* 2009; 24:89–100
12. Wiedemann HP, Wheeler AP, Bernard GR, et al: Comparison of two fluid-management strategies in acute lung injury. *N Engl J Med* 2006; 354:2564–2575
13. Thiel M, Chouker A, Ohta A, et al: Oxygenation inhibits the physiological tissue-protecting mechanism and thereby exacerbates acute inflammatory lung injury. *PLoS Biol* 2005; 3:e174
14. Jackson RM: Molecular, pharmacologic, and clinical aspects of oxygen-induced lung injury. *Clin Chest Med* 1990; 11:73–86
15. Martin DS, Grocott MP: Oxygen therapy in critical illness: Precise control of arterial oxygenation and permissive hypoxemia. *Crit Care Med* 2013; 41:423–432
16. Yapicioğlu H, Yildizdaş D, Bayram I, et al: The use of surfactant in children with acute respiratory distress syndrome: Efficacy in terms of oxygenation, ventilation and mortality. *Pulm Pharmacol Ther* 2003; 16:327–333
17. Willson DF, Thomas NJ: Surfactant composition and biophysical properties are important in clinical studies. *Am J Respir Crit Care Med* 2010; 181:762; author reply 763
18. Willson DF, Thomas NJ, Markovitz BP, et al; Pediatric Acute Lung Injury and Sepsis Investigators: Effect of exogenous surfactant (calfactant) in pediatric acute lung injury: A randomized controlled trial. *JAMA* 2005; 293:470–476
19. Willson DF, Thomas NJ, Tamburro R, et al; Pediatric Acute Lung and Sepsis Investigators Network: Pediatric calfactant in acute respiratory distress syndrome trial. *Pediatr Crit Care Med* 2013; 14:657–665
20. Papazian L, Forel JM, Gacouin A, et al; ACURASYS Study Investigators: Neuromuscular blockers in early acute respiratory distress syndrome. *N Engl J Med* 2010; 363:1107–1116
21. Alhazzani W, Alshahrani M, Jaeschke R, et al: Neuromuscular blocking agents in acute respiratory distress syndrome: A systematic review and meta-analysis of randomized controlled trials. *Crit Care* 2013; 17:R43
22. Hall JB: Point: Should paralytic agents be routinely used in severe ARDS? Yes. *Chest* 2013; 144:1440–1442
23. Sessler CN: Counterpoint: Should paralytic agents be routinely used in severe ARDS? No. *Chest* 2013; 144:1442–1445
24. Randolph AG: Management of acute lung injury and acute respiratory distress syndrome in children. *Crit Care Med* 2009; 37:2448–2454
25. Afshari A, Brok J, Moller AM, et al: Inhaled nitric oxide for acute respiratory distress syndrome (ARDS) and acute lung injury in children and adults. *Cochrane Database Syst Rev* 2010; 7:CD002787
26. Adhikari NK, Burns KE, Friedrich JO, et al: Effect of nitric oxide on oxygenation and mortality in acute lung injury: Systematic review and meta-analysis. *BMJ* 2007; 334:779

27. Meduri GU, Golden E, Freire AX, et al: Methylprednisolone infusion in early severe ARDS: Results of a randomized controlled trial. *Chest* 2007; 131:954–963
28. Meduri GU, Marik PE, Chrousos GP, et al: Steroid treatment in ARDS: A critical appraisal of the ARDS network trial and the recent literature. *Intensive Care Med* 2008; 34:61–69
29. Tang BM, Craig JC, Eslick GD, et al: Use of corticosteroids in acute lung injury and acute respiratory distress syndrome: A systematic review and meta-analysis. *Crit Care Med* 2009; 37:1594–1603
30. Meduri GU, Annane D, Chrousos GP, et al: Activation and regulation of systemic inflammation in ARDS: Rationale for prolonged glucocorticoid therapy. *Chest* 2009; 136:1631–1643
31. Ware LB, Matthay MA: The acute respiratory distress syndrome. *N Engl J Med* 2000; 342:1334–1349
32. Meduri GU, Muthiah MP, Carratu P, et al: Nuclear factor-kappaB- and glucocorticoid receptor alpha-mediated mechanisms in the regulation of systemic and pulmonary inflammation during sepsis and acute respiratory distress syndrome. Evidence for inflammation-induced target tissue resistance to glucocorticoids. *Neuroimmunomodulation* 2005; 12:321–338
33. Confalonieri M, Urbino R, Potena A, et al: Hydrocortisone infusion for severe community-acquired pneumonia: A preliminary randomized study. *Am J Respir Crit Care Med* 2005; 171:242–248
34. Sabry NA, Omar EE-D: Corticosteroids and ICU course of community acquired pneumonia in Egyptian settings. *Pharmacol Pharm* 2011; 2:73–81
35. Annane DS,Sébille V, Bellissant E, for the Ger-Inf-05 Study Group: Effect of low doses of corticosteroids in septic shock patients with or without early acute respiratory distress syndrome. *Crit Care Med* 2006; 34:22–30
36. Meduri GU, Golden E, Freire AX, et al: Methylprednisolone infusion in early severe ARDS: Results of a randomized controlled trial. *Chest* 2007; 131:954–963
37. Rezk N, Ibrahim A: Effects of methylprednisolone in early ARDS. *Egypt J Chest Dis Tuberc* 2013; 62:167–172
38. Meduri GU, Headley AS, Golden E, et al: Effect of prolonged methylprednisolone therapy in unresolving acute respiratory distress syndrome: A randomized controlled trial. *JAMA* 1998; 280:159–165
39. Steinberg KP, Hudson LD, Goodman RB, et al; National Heart, Lung, and Blood Institute Acute Respiratory Distress Syndrome (ARDS) Clinical Trials Network: Efficacy and safety of corticosteroids for persistent acute respiratory distress syndrome. *N Engl J Med* 2006; 354:1671–1684
40. Martinot A, Fourier C, Cremer R, et al: Short-course, high-dose corticosteroid treatment in six children with late ARDS. *Pediatr Pulmonol* 1997; 23:314–316
41. Goh AY, Sekaran D, Roziah M: Corticosteroid rescue in late paediatric acute respiratory distress syndrome. *Respirology* 1999; 4:295–297
42. Guglani L, Jain S, Lodha R: Methylprednisolone therapy in a child with unresolving ARDS. *Indian Pediatr* 2006; 43:639–642
43. De Luca D, Minucci A, Cogo P, et al: Secretory phospholipase A_2 pathway during pediatric acute respiratory distress syndrome: A preliminary study. *Pediatr Crit Care Med* 2011; 12:e20–e24
44. Marshall RP, Webb S, Bellingan GJ, et al: Angiotensin converting enzyme insertion/deletion polymorphism is associated with susceptibility and outcome in acute respiratory distress syndrome. *Am J Respir Crit Care Med* 2002; 166:646–650
45. Plunkett A, Agbeko RS, Li K, et al: Angiotensin-converting enzyme D allele does not influence susceptibility to acute hypoxic respiratory failure in children. *Intensive Care Med* 2008; 34:2279–2283
46. Honiden S, Gong MN: Diabetes, insulin, and development of acute lung injury. *Crit Care Med* 2009; 37:2455–2464
47. Burri PH: Structural aspects of postnatal lung development–Alveolar formation and growth. *Biol Neonate* 2006; 89:313–322
48. Smith LS, Zimmerman JJ, Martin TR: Mechanisms of acute respiratory distress syndrome in children and adults: A review and suggestions for future research. *Pediatr Crit Care Med* 2013; 14:631–643
49. Bernard GR, Artigas A, Brigham KL, et al: Report of the American-European consensus conference on ARDS: Definitions, mechanisms, relevant outcomes and clinical trial coordination. The Consensus Committee. *Intensive Care Med* 1994; 20:225–232
50. Altman DG, Schulz KF, Moher D, et al; CONSORT GROUP (Consolidated Standards of Reporting Trials): The revised CONSORT statement for reporting randomized trials: Explanation and elaboration. *Ann Intern Med* 2001; 134:663–694
51. Raurich JM, Ferreruela M, Llompart-Pou JA, et al: Potential effects of corticosteroids on physiological dead-space fraction in acute respiratory distress syndrome. *Respir Care* 2012; 57:377–383
52. Tang B, Craig JC, Eslick GD, et al: Use of corticosteroids in acute lung injury and acute respiratory distress syndrome: A systematic review and meta-analysis. *Crit Care Med* 2009; 37:1594–1603
53. Slutsky AS, Ranieri VM: Ventilator-induced lung injury. *N Engl J Med* 2013; 369:2126–2136
54. Bernard GR, Luce JM, Sprung CL, et al: High-dose corticosteroids in patients with the adult respiratory distress syndrome. *N Engl J Med* 1987; 317:1565–1570
55. Khemani RG, Newth CJ: The design of future pediatric mechanical ventilation trials for acute lung injury. *Am J Respir Crit Care Med* 2010; 182:1465–1474
56. Khemani RG, Wilson DF, Esteban A, et al: Evaluating the Berlin Definition in pediatric ARDS. *Intensive Care Med* 2013; 39:2213–2216